

3. To create and manipulate datasets using Pandas library on the Iris dataset.

```
In [1]: import pandas as pd

# Create a DataFrame

data = {
    'Name': ['Amit', 'Priya', 'Rahul'],
    'Age': [20, 21, 22],
    'Marks': [85, 90, 88]
}
df = pd.DataFrame(data)
```

```
In [2]: # Display DataFrame

print("DataFrame:")
print(df)
```

DataFrame:

	Name	Age	Marks
0	Amit	20	85
1	Priya	21	90
2	Rahul	22	88

```
In [3]: # Display first two rows

print("\nFirst Two Rows:")
print(df.head(2))
```

First Two Rows:

	Name	Age	Marks
0	Amit	20	85
1	Priya	21	90

```
In [4]: # Display column names

print("\nColumns:")
print(df.columns)
```

Columns:

Index(['Name', 'Age', 'Marks'], dtype='object')

```
In [5]: # Display shape

print("\nShape:")
print(df.shape)
```

Shape:

(3, 3)

In [7]: *#Load Iris Dataset from CSV File*

```
import pandas as pd

# Read Iris CSV file
df = pd.read_csv(r"C:\Users\Smit\OneDrive\Desktop\Dataset\IRIS.csv")

# Display first 5 rows
print("First 5 Rows:")
print(df.head())

# Display dataset information
print("\nDataset Information:")
print(df.info())

# Display statistical summary
print("\nStatistical Summary:")
print(df.describe())

# Display shape of dataset
print("\nShape of Dataset:")
print(df.shape)
```

First 5 Rows:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 150 entries, 0 to 149

Data columns (total 5 columns):

#	Column	Non-Null Count	Dtype
0	sepal_length	150 non-null	float64
1	sepal_width	150 non-null	float64
2	petal_length	150 non-null	float64
3	petal_width	150 non-null	float64
4	species	150 non-null	object

dtypes: float64(4), object(1)

memory usage: 6.0+ KB

None

Statistical Summary:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

Shape of Dataset:

(150, 5)

In [8]: *#Basic Data Manipulation on Iris Dataset*

```
import pandas as pd

# Load Iris Dataset
df = pd.read_csv(r"C:\Users\Smit\OneDrive\Desktop\Dataset\IRIS.csv")

# Display column names
print("Columns:")
print(df.columns)

# Filter records where sepal_length > 5.0
filtered_data = df[df["sepal_length"] > 5.0]

print("\nFiltered Data:")
print(filtered_data.head())

# Average Sepal Length
print("\nAverage Sepal Length:")
print(df["sepal_length"].mean())

# Count flowers by species
print("\nSpecies Count:")
print(df["species"].value_counts())
```

Columns:
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
 'species'],
 dtype='object')

Filtered Data:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
5	5.4	3.9	1.7	0.4	Iris-setosa
10	5.4	3.7	1.5	0.2	Iris-setosa
14	5.8	4.0	1.2	0.2	Iris-setosa
15	5.7	4.4	1.5	0.4	Iris-setosa

Average Sepal Length:
5.843333333333334

Species Count:
Iris-setosa 50
Iris-versicolor 50
Iris-virginica 50
Name: species, dtype: int64

In [10]: *# Practical 3: Pandas Operations on Iris Dataset*

```
import pandas as pd

# Load the Iris Dataset
df = pd.read_csv(r"C:\Users\Smit\OneDrive\Desktop\Dataset\IRIS.csv")

# Display the first 5 records
print("First 5 Records:")
print(df.head())

# Display the last 5 records
print("\nLast 5 Records:")
print(df.tail())

# Display dataset information
print("\nDataset Information:")
print(df.info())

# Display the shape of the dataset
print("\nShape of Dataset:")
print(df.shape)

# Display column names
print("\nColumn Names:")
print(df.columns)

# Display summary statistics
print("\nSummary Statistics:")
print(df.describe())

# Check for missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Display unique species
print("\nUnique Species:")
print(df["species"].unique())

# Count records of each species
print("\nSpecies Count:")
print(df["species"].value_counts())

# Select specific columns
print("\nSepal Length and Species:")
print(df[["sepal_length", "species"]].head())

# Filter rows (Sepal Length > 5.0)
print("\nRows where Sepal Length > 5.0:")
print(df[df["sepal_length"] > 5.0])

# Sort data by Petal Length
print("\nSorted by Petal Length:")
print(df.sort_values(by="petal_length"))

# Add a new column
df["Sepal_Area"] = df["sepal_length"] * df["sepal_width"]

print("\nDataset with New Column:")
print(df.head())
```

```
# Save the modified dataset
df.to_csv("Modified_IRIS.csv", index=False)

print("\nModified dataset saved successfully.")
```

First 5 Records:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Last 5 Records:

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 150 entries, 0 to 149
```

```
Data columns (total 5 columns):
```

In []: